**MINISTRY OF EDUCATION AND TRAINING**
**CAN THO UNIVERSITY**

# SUMMARY REPORT RESEARCH PROJECT

- Project code: THS2026-68
- Project title: Development of an Edge AI-based System for Multimodal Analysis and Clustering of Flood Rescue Events
- Project period: May–October 2026
- Total cost: 15,000,000 VND
- Project leader: Lê Thị Ngọc Ảnh (0376932994)
- Project members: Nguyễn Thanh Trọng, Cao Tường Hưng, Nguyễn Như Quỳnh, Ngô Hưng Thịnh.
- Advisor: TS. Nguyễn Thanh Khoa (ntkhoa@ctu.edu.vn)
- Necessity of the project: Duplicate, unlocated or delayed rescue reports hinder coordination under intermittent connectivity.
- Objectives: Develop a prototype combining edge image inference, compact transmission and map-based event clustering.
- Methodology: Train MobileNetV3-Large and evaluate image classification, clustering, ranking and dispatch separately using saved artifacts and simulation datasets.
- Project activities: Image AI, structured urgency, persistent queues, adaptive sending, synchronization and dashboard. A translated-Vietnamese/AI-silver pilot runs NLP and offline fusion; native data, information extraction and camera/Android evaluation remain open.
- Research results: The baseline reaches 75.78% accuracy and 74.26% macro-F1 on 256 images, with three critical errors; phone benchmarks cover 32 configurations on four devices. McNemar does not establish historical pruning gains; a 5/10/20% grid does not preserve quality. PTQ/QAT diagnostics remain partial. The app, backend and dashboard exist; the ranking adapter does not use LR scores. English CrisisMMD fusion reaches 78.35% macro-F1 with effect CIs including zero. On team-checked Vietnamese translations/AI-silver labels, fusion reaches 64.91% macro-F1 and 33.33% request recall; CUDA T2 averages 52.29% macro-F1. The team confirms mobile NLP/fusion integration and that the offline-submission issue has been fixed.
- Research new finding: RQ1 does not establish Product superiority; RQ2 does not establish general alignment gains and retains three raw Spearman and five Holm replay differences. RQ3 favors nearest-first over revised priority on simulated harm and deadline misses. Field rescue benefits remain unmeasured.
- Application potentials: Research and workflow demonstration. Urgency integration, mobile provenance, repeated device/network/SMS tests and expert validation remain necessary.